OpenIoT: Enabling the Convergence *of* IoT *and* Cloud Computing

OpenIoT is a simple-to-use open source application, which connects all the sensors that use cloud technologies to make them an extension of your IoT application.

Everyone knows that the Internet of Things (IoT) is competing with other technologies due to its platform-independent and robust characteristics. More importantly, IoT now assists the cloud computing platform due to its scalability and easy access to sophisticated hardware.

This convergence of IoT and cloud architecture has not only dominated the research domain but also the enterprise sector, for the past few years. The thing that makes these two completely versatile ideas compatible with each other is their capability to simultaneously manage the application, the user interfaces and the data stream, especially in a high performance support structure. However, these combined structures might not necessarily support semantic interoperability across IoT implementations, which are mainly required to be sustainably independent from each other. Therefore, it might not be valid to decide which of the approaches is the most efficient in order to bring together data streams and services provided by diverse IoT platforms that feature incompatible semantics due to cross-platform compatibility (e.g., units of measurement, raw sensor values and points of interest).

OpenIoT

Open IoT bridges this gap between the semantics and data computing to form a unified semantics for IoT apps in cloud interfaces. Open IoT uses the W3C Semantic Sensor Networks (SSN) ontology as a generalised standard model for semantic unification of diverse IoT systems. It offers versatility in its infrastructure for gathering and semantically annotating data from any I/O device available. Open IoT also has a special feature that can easily help to link data sets with each other, depending upon their similarities. As a consequence, it can dynamically deal with the specific